Task 1 | Related Work: Five-Paper Comparison

Our data: BCI Competition IV-2a - 9 participants, 22 EEG channels, 4 motor-imagery classes (left hand, right hand, feet, tongue), sampled at 250 Hz.

Paper (year)	Dataset / area	Method / GNN type	Graph construction	Accuracy	Strength	Limitation	Research gap	Relation to our project
Sun et al. (2023) EEG Graph Self-Attention	Two MI-EEG datasets Motor imagery BCI	EEG Graph Self-Attention Network (EGSAN)	EEG-channel spatial graph; attention learns channel importance	Not confirmed	Learns useful graph features	Limited comparison of graph designs	Test alternative connectivity graphs	Compare distance and functional channel graphs
Liang et al. (2023) GCN Channel Selection	Three MI-EEG datasets Channel selection	Graph Convolutional Network (GCN)	Pearson correlation between channels	79.76%, 89.14%, 87.96% (different datasets)	Identifies useful electrodes	Mainly focuses on channel selection	Test accuracy with fewer channels	Evaluate channel efficiency on IV-2a
Shi et al. (2024) Brain Topography Graph-CNN	BCI IV-2a; HGD MI classification	GCN + CNN + TCN	Mutual-information graph	80.46% on IV-2a	Learns spatial and temporal patterns	Complex hybrid model	Test simpler graph models fairly	Relevant IV-2a benchmark
Zhu et al. (2023) Multi-Scale Attention GCN	MI-EEG Exact dataset unconfirmed	GCN + multi-scale attention	Phase Lag Index (PLI) connectivity	93.22% (10-fold CV)	Captures functional links	Protocol not verified for IV-2a	Compare connectivity under one split	Motivates functional graph testing
Abdelaty et al. (2025) Graph Vertex & Spectral Features	BCI IV-2a; Dataset 1 MI classification	Graph features + conventional ML (not GNN)	Weighted, directed EEG graph	81.75% on IV-2a	Interpretable graph features	Handcrafted features	Compare handcrafted vs learned features	Graph-feature baseline on IV-2a

Research gap: Existing studies use different EEG graphs, features and evaluation methods. Our project will compare distance-based and functional connectivity graphs, feature types, and channel efficiency on BCI IV-2a using consistent, leakage-free evaluation.

Note: Accuracies from different datasets or validation protocols are not directly comparable. A missing value is marked as unconfirmed rather than estimated.